Tables

Table I. Primary functions in whole-firm recruitment

Primary function	Ads	Ad-weighted %	Firm-equal %
Manufacturing, quality and supply chain	214	13.1	13.0
Software and platforms	180	11.0	10.8
ML/AI engineering	156	9.6	9.7
Corporate support	132	8.1	6.7
Hardware and electrical engineering	123	7.5	8.2
Commercial and customer-facing work	117	7.2	7.3
Product and program management	106	6.5	5.9
Mechanical and electromechanical engineering	89	5.5	5.6
Deployment and field service	84	5.2	5.2
Control and planning	75	4.6	6.2
Data and analytics	70	4.3	4.5
Simulation, evaluation and measurement toolchains	60	3.7	3.3
Perception	47	2.9	2.6
Systems engineering and cross-domain integration	31	1.9	2.1
Industrial and experience design	23	1.4	1.3
Research science	23	1.4	1.1
Robot operations	22	1.3	2.7
Composite and multi-direction recruitment	30	1.8	1.6
Ambiguous, unmapped or insufficient-source	48	2.9	2.5

Notes: Ad-weighted percentages use all 1,630 whole-firm advertisements as the denominator: 1,552 advertisements assigned to specific functions and 78 structural or unresolved records. Firm-equal percentages average the within-firm shares. Source: calculations from the job-advertisement corpus.

Table II. Activity within functions and the distribution of activity advertisements

Activity	Primary function	Activity / function N	Within function %	Activity / total activity	Activity distribution %
Research activity	Research science	18/23	78.3	18/66	27.3
Research activity	ML/AI engineering	36/156	23.1	36/66	54.5
Research activity	Perception	5/47	10.6	5/66	7.6
Quality-related activity	Manufacturing, quality and supply chain	159/214	74.3	159/459	34.6
Quality-related activity	Hardware and electrical engineering	53/123	43.1	53/459	11.5
Quality-related activity	Mechanical and electromechanical engineering	36/89	40.4	36/459	7.8
Quality-related activity	Deployment and field service	34/84	40.5	34/459	7.4
Quality-related activity	Systems engineering and integration	21/31	67.7	21/459	4.6

Notes: Function counts are limited to advertisements for which activity status could be evaluated. Research activity appears in 66 advertisements and quality-related activity in 459. The text reports the main contributing functions; activity_bidirectional.csv provides every category, including zeros, structural categories and unknowns. Research science and manufacturing, quality and supply chain were prespecified as the corresponding functions; they were not selected from the results. Source: calculations from the job-advertisement corpus.

Table III, Panel A. Engineering-activity decomposition across sensitivity scenarios

Scenario	Galbot coverage	Figure coverage	Total gap pp	Composition pp	Within function pp
Main scope	82/181	60/126	+11.10	+19.45	-8.35
Exclude campus and internship ads	63/126	60/122	+10.56	+21.08	-10.52
Deduplicate duty templates	79/177	59/118	+12.89	+19.43	-6.54

Notes: Galbot minus Figure, within the fixed set of four common functions. Coverage denominators are advertisements for which activity status could be evaluated in each scenario. Supplementary Table S7 reports the complete scenarios and missing comparisons. pp denotes percentage points. Source: calculations from the job-advertisement corpus.

Table III, Panel B. Function counts and components of the primary decomposition

Primary function	Galbot activity/ads	Figure activity/ads	Galbot/Figure shares %	Composition pp	Within function pp
Corporate support	0/13	1/11	15.9 / 18.3	-0.11	-1.55
Manufacturing, quality and supply chain	3/20	5/28	24.4 / 46.7	-3.66	-1.02
ML/AI engineering	20/24	10/10	29.3 / 16.7	+11.55	-3.83
Software and platforms	23/25	11/11	30.5 / 18.3	+11.67	-1.95

Notes: The common-scope denominators are 82 for Galbot and 60 for Figure. Activity counts / advertisements recover the within-function engineering-activity rates. The composition and within-function components sum to +19.45 and −8.35 percentage points, respectively. Source: calculations from the job-advertisement corpus.

Table IV, Panel A. Technology mentions and statement types

Task and technology	M positive/N	Ad-weighted %	Firm-equal %	Duties	Required	General	Preferred
Robot modeling — Python	73/105	69.5	66.9	3	2	65	6
Robot modeling — PyTorch	83/105	79.0	74.5	1	4	75	4
Robot runtime software — C++	80/87	92.0	91.4	13	1	73	6
Robot runtime software — Linux	52/87	59.8	50.2	9	0	50	6
Board-level circuitry — Altium	24/56	42.9	35.2	5	0	21	5
Board-level circuitry — EMC / SI / PI	36/56	64.3	67.8	27	0	20	11

Notes: The last four columns count statement types among advertisements with a mention; each count can be divided by the M-positive count in that row, and types may overlap. DUTY combines the focal person's use and development statements at the advertisement level. Firm-equal means include firms with a positive valid denominator and zero positives. EMC/SI/PI is a family indicator for compatibility, integrity and related protection methods; its members are unioned and deduplicated by advertisement. It is therefore broader than the single Altium tool. Supplementary Section S1 provides family members and branching rules. Source: calculations from the job-advertisement corpus.

Table IV, Panel B. Unconditional statements in general qualifications

Task and technology	G positive/N	U	Ad-weighted %	Firm-equal %	Unknown bounds %
Robot modeling — Python	40/88	17	45.5	41.7	38.1 to 54.3
Robot modeling — PyTorch	20/97	8	20.6	20.4	19.0 to 26.7
Robot runtime software — C++	38/64	23	59.4	51.5	43.7 to 70.1
Robot runtime software — Linux	47/86	1	54.7	46.0	54.0 to 55.2
Board-level circuitry — Altium	5/56	0	8.9	8.8	8.9 to 8.9
Board-level circuitry — EMC / SI / PI	15/56	0	26.8	28.2	26.8 to 26.8

Notes: G here displays only GENERAL—UNCONDITIONAL and does not combine explicitly REQUIRED statements. The point estimate is P/N, where N=P+Z. Unknown-assignment bounds run from P/B to (P+U)/B, where B=N+U; they are not confidence intervals. All 22 qualification-relation cells are provided in the supplementary data. Source: calculations from the job-advertisement corpus.

Table V. General-qualification context among technology mentions

Task and technology	Mentions	G positive P	P/mentions %	No G positive Z	G unknown U	Choice/example in Z+U
Robot modeling — Python	73	40	54.8	16	17	11/33
Robot modeling — PyTorch	83	20	24.1	55	8	52/63
Robot runtime software — C++	80	38	47.5	19	23	16/42
Robot runtime software — Linux	52	47	90.4	4	1	2/5
Board-level circuitry — Altium	24	5	20.8	19	0	17/19
Board-level circuitry — EMC / SI / PI	36	15	41.7	21	0	6/21

Notes: P, Z and U are mutually exclusive among mention advertisements and sum to the mention count. P / mentions uses all mention advertisements in the row as its denominator and retains U; the Table IV, Panel B point estimate uses P+Z for the relevant context within the task. In the last column, the denominator is Z+U and the numerator counts advertisements containing at least one choice expression, example expression, or both. These expressions may overlap with PREFERRED, DUTY, REQUIRED and unresolved branches, so this column is not an exhaustive mutually exclusive decomposition. Source: calculations from the job-advertisement corpus.